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DESIGN LABORATORY - II (ME306) PROJECT REPORT

For

Wave Energy Converter Control System with Vibration and Mechanical Control

Submitted by

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1. Title :

Wave Energy Converter Control System with Vibration and Mechanical Control

2. Objective :

The objective of this project is to design and simulate a Wave Energy Converter (WEC) system capable of efficiently harvesting energy from ocean waves. The focus is on developing a vibration-based control mechanism that optimizes energy conversion by minimizing mechanical vibrations while maximizing power output thereby improving performance in dynamic and varying wave conditions. This is achieved through a by tuning the PID controller parameters using a data-driven machine learning approach. The system is modelled in Simulink and Simscape, aiming to provide a sustainable and adaptive solution under varying ocean wave conditions.

3. Short Description :

This project focuses on the development of a Wave Energy Converter (WEC) system designed to extract renewable energy from ocean waves. The system is modeled as a mass-spring-damper structure subjected to wave-induced forces, with the goal of optimizing energy conversion efficiency. A Proportional-Integral-Derivative (PID) controller is implemented to regulate the dynamic response of the system, while minimizing mechanical vibrations. To further enhance performance, a machine learning (ML)-based regression model is employed to predict optimal PID parameters that maximize power output. The modeling and simulations are carried out using MATLAB Simulink and Simscape, ensuring a high-fidelity representation of the physical dynamics. This integrated approach combines control engineering, data-driven optimization, and sustainable energy harvesting, offering a robust solution adaptable to varying sea conditions.

4. System Simulation using Simulink and Simscape :

4.1 Model Architecture :

The Wave Energy Converter (WEC) system is modelled in MATLAB Simulink and Simscape to closely replicate the physical dynamics of a buoy-type energy harvesting device subjected to ocean wave forces. The mechanical behavior of the system is represented as a **mass-spring-damper model**, which effectively captures the vibration and displacement characteristics of the floating body in response to wave excitation.

The key components of the model include e:

- **Wave Input Signal:** A sinusoidal signal representing periodic ocean wave motion is used as the input. This signal is mathematically modelled as $F(t) = A \backslash sin(\omega t)$, where A is the amplitude and ω is the angular frequency.
- **Force Conversion Block:** The wave input is converted into a mechanical force acting on the mass (buoy), simulating the hydrodynamic interaction.
- **Mass-Spring-Damper System:** This models the physical buoy system, where:
  - *Mass* represents the effective mass of the floating body.

- *Spring* represents the restoring force due to buoyancy and mooring.
- *Damper* represents hydrodynamic damping or internal mechanical losses.

$$m\frac{dx^2}{dt^2} + c\frac{dx}{dt} + k\,x = F$$

- **PID Controller:** A closed-loop feedback controller is implemented to tune the input signal or system response, with parameters Kp, Ki, and Kd. The PID modifies the system's behavior in real time to optimize energy extraction.

$$\boldsymbol{u(t) = K_p e(t) + K_i \int e(t)dt + \frac{K_d de(t)}{dt}}$$

- **Measurement Blocks:** Sensors are modelled to measure key outputs such as displacement, velocity, and the resulting **power output**, calculated as the product of velocity and applied force:

$$Power(t) = F(t) \cdot v(t)$$

4.2 Simulation Workflow

The simulation process is structured into the following stages:

1. **Initialization:** Parameter values for the mass, damping coefficient, spring constant, and wave properties are defined.
2. **Wave Interaction Modelling:** The sinusoidal force input is applied to the system and the resulting motion is captured.
3. **Control Implementation:** The PID controller adjusts the system's response by minimizing vibrations and improving dynamic tracking.
4. **Data Collection:** For each simulation, the system's velocity, displacement, and instantaneous power output are recorded.
5. **PID Parameter Sweep:** To generate training data for machine learning, a systematic variation of Kp, Ki, and Kd values is performed. A total of **250 simulations** were run, each with a unique set of PID parameters.
6. **Exporting Simulation Results:** The resulting dataset—including PID parameters and corresponding power outputs—is used as input for the machine learning model.

This simulation setup not only reflects the physics of the WEC system but also enables rapid testing and optimization of control strategies without the need for a physical prototype. By integrating Simscape's physics-based components with Simulink's control and signal processing capabilities, a robust and scalable simulation environment is achieved.

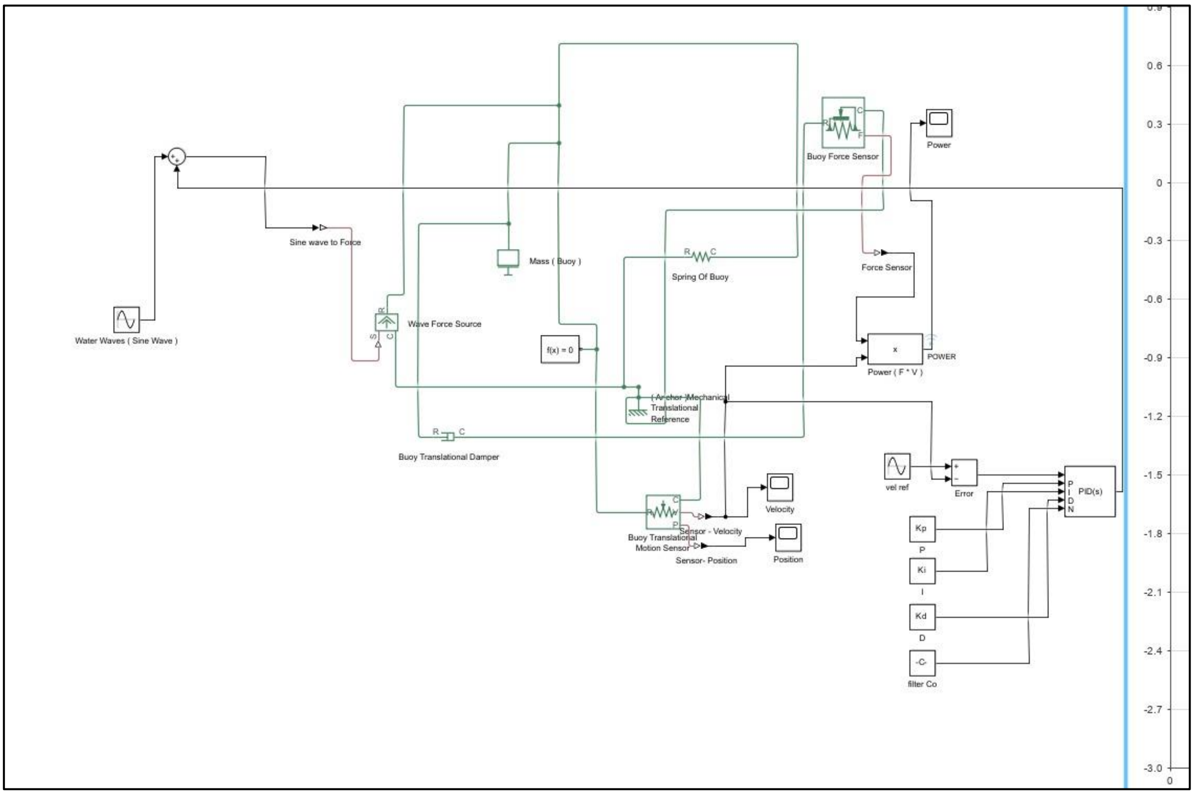


Fig - Simulink model of WEC buoy & PID controller configuration

5. Machine Learning Model Design :

5.1 Problem Framing :

A regression model was built to predict the power output of the WEC system for a given set of PID parameters:

Input: (Kp, Ki, Kd)
Output: Predicted system power

5.2 Model Architecture

- A **feedforward neural network** was implemented with:
  - **Input layer:** 3 neurons for PID values.
  - **Hidden layers:** ReLU activation.
  - **Output layer:** 1 neuron for predicted power.

5.3 Training and Validation

- 250 data points were used for training and validation.
- The model was trained using **Adam optimizer** and **backpropagation**.
- Loss was defined as **mean squared error (MSE)**.
- Training was conducted over **400 epochs**.

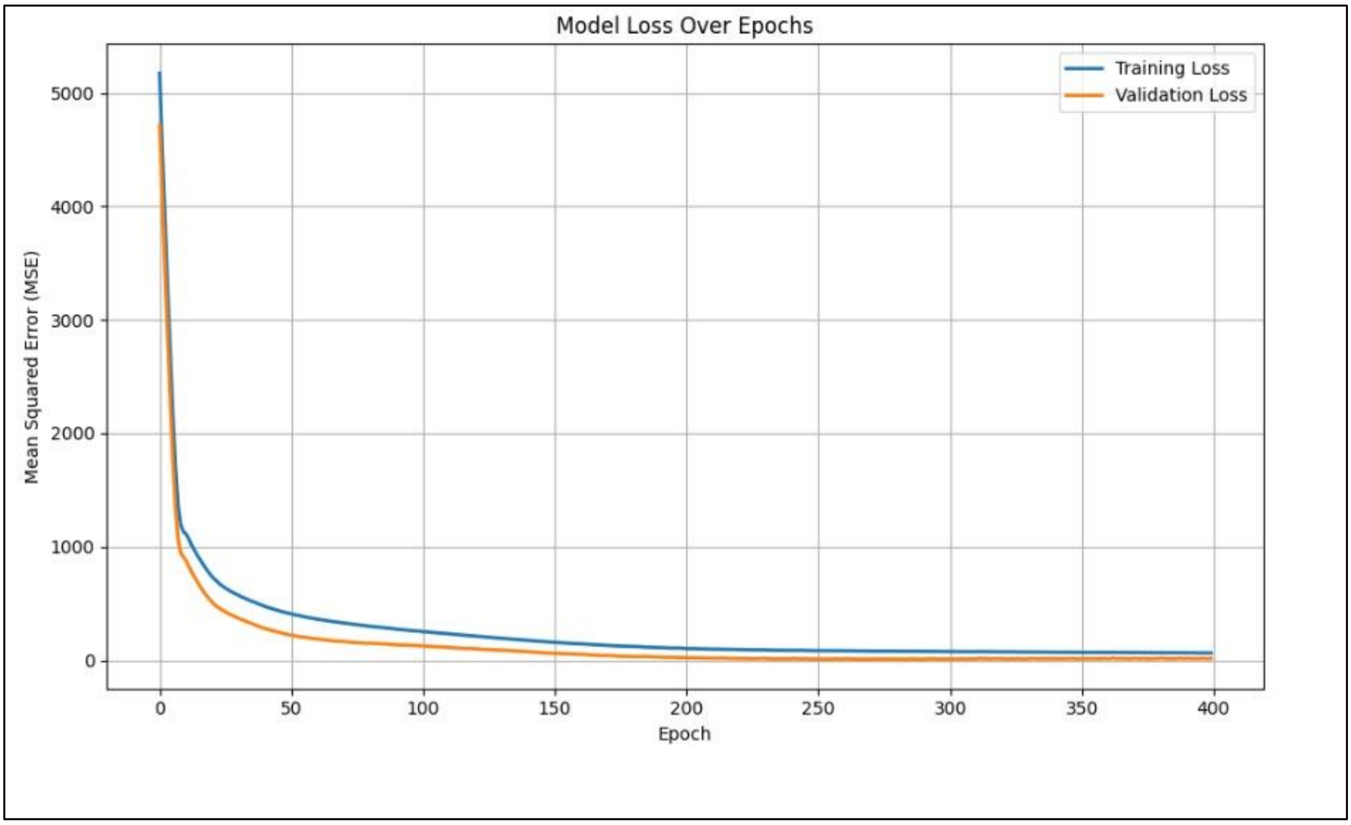


Fig - Loss curves show a steady convergence of training and validation errors.

6. Optimization Process :

6.1 Predicting Optimal PID Values :

- A synthetic dataset of **500,000 combinations** of (Kp, Ki, Kd) was generated.
- The ML model predicted power output for each.
- Optimal parameters were chosen based on **maximum predicted power**.

6.2 Best Parameter Selection :

- Best performance observed at :
  - **Kp = 28**, **Ki = 0**, **Kd = 12**
  - **Maximum Power = 180 W**

7. Comparison with Other Methods

Method	Kp	Ki	Kd	Output Power (W)
ML-Optimized	28	0	12	180
Manual Trial	1	0	0	30

Method	Kp	Ki	Kd	Output Power (W)
Simulink PID Tuner	-	-	-	160

8. Advantages of the Approach

- **No physical testing** needed during tuning.
- **Extremely fast prediction** after training.
- **Scalable** to different systems or objectives (vibration damping, energy maximization, etc.).
- Supports **continuous improvement** through retraining on new data.

9. Visuals & Results :

➢ **Wave energy converter simulation plots:**

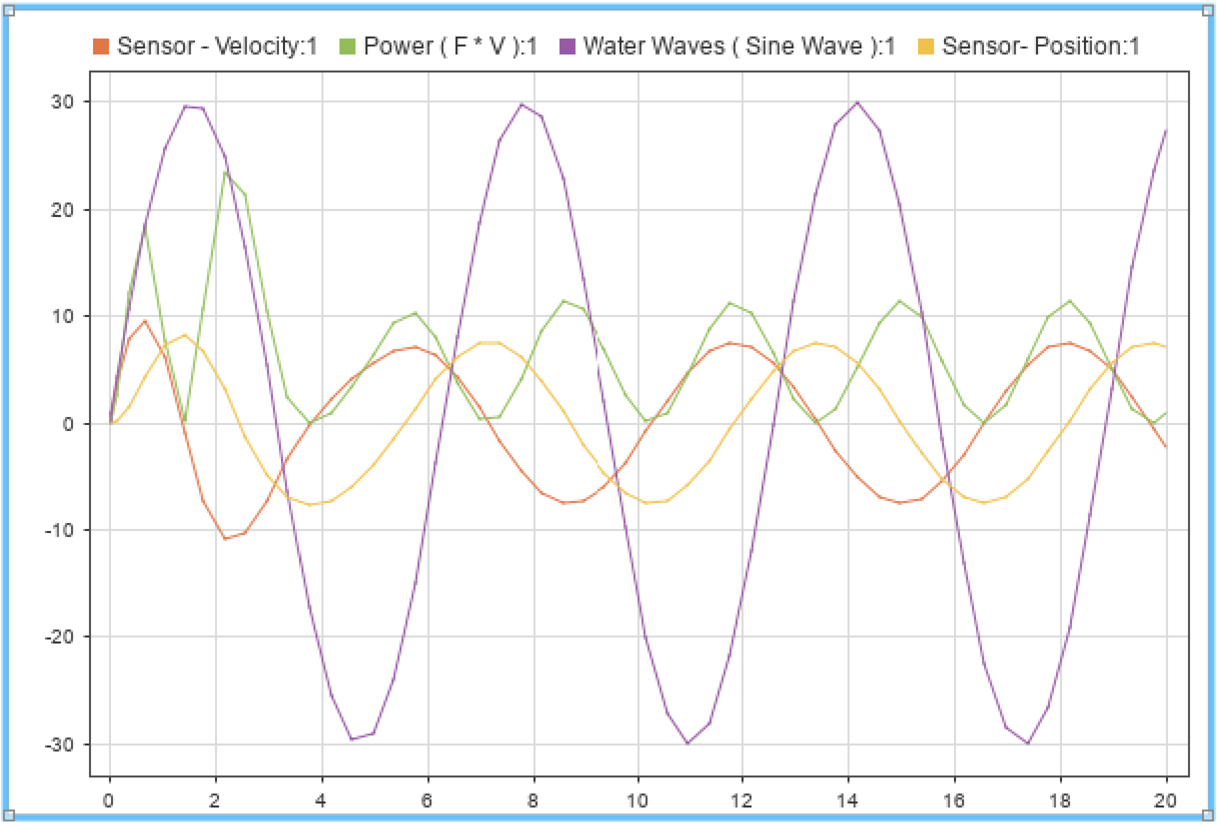


Fig - Simulation results (Power, Velocity , Position) in initial PID conditions , Kp = 1 , Ki = 0 , Kd = 0 .

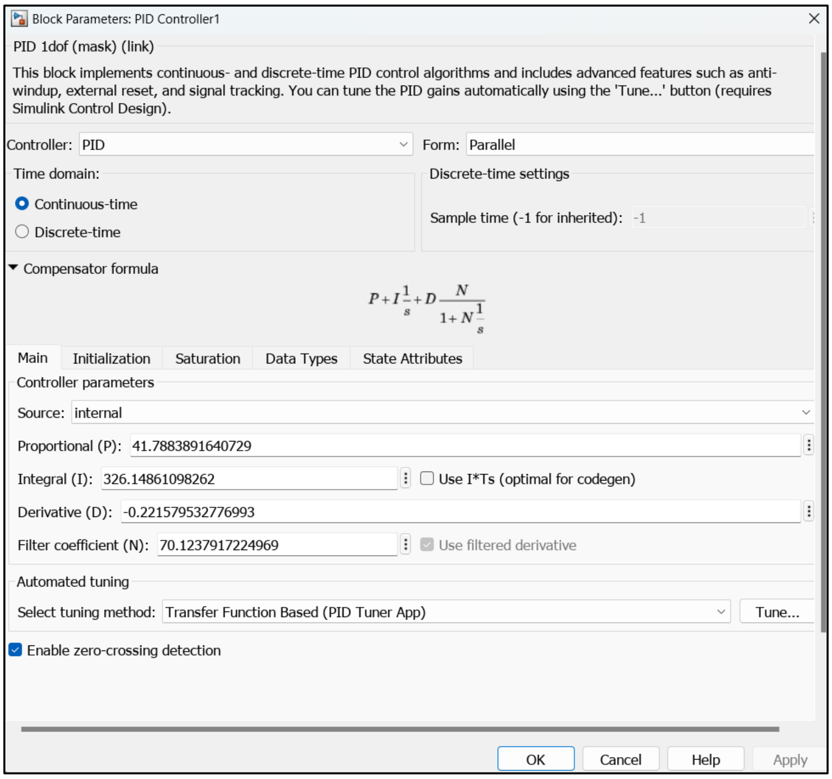


Fig - PID Parameters when auto tuned using Transfer Function Based (PID Tuner App) in simulink for first order transfer function.

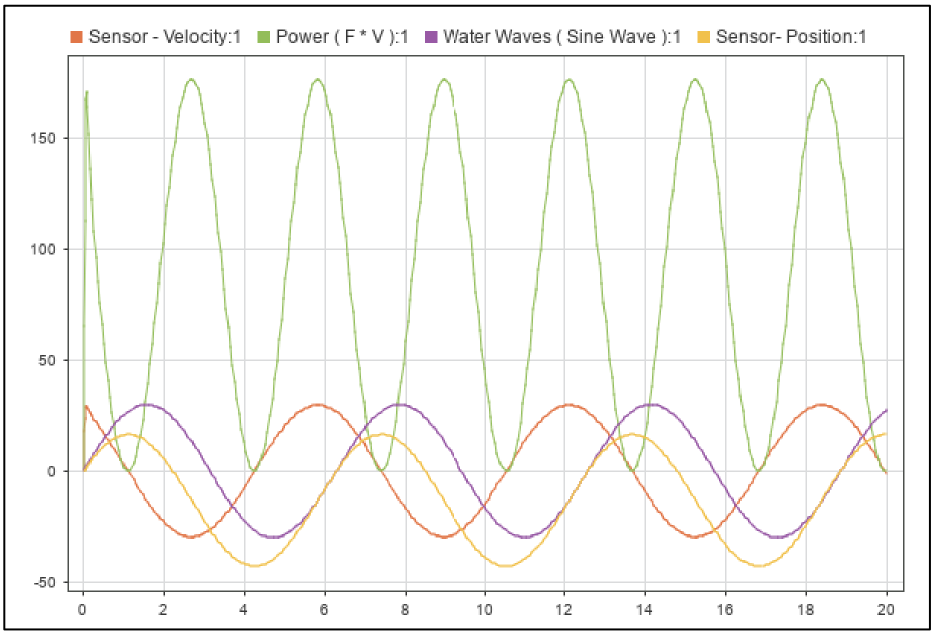


Fig - Simulation results (Power, Velocity , Position) in auto tuned PID conditions , Kp = 41.788 , Ki = 326.148, Kd = -0.2215.

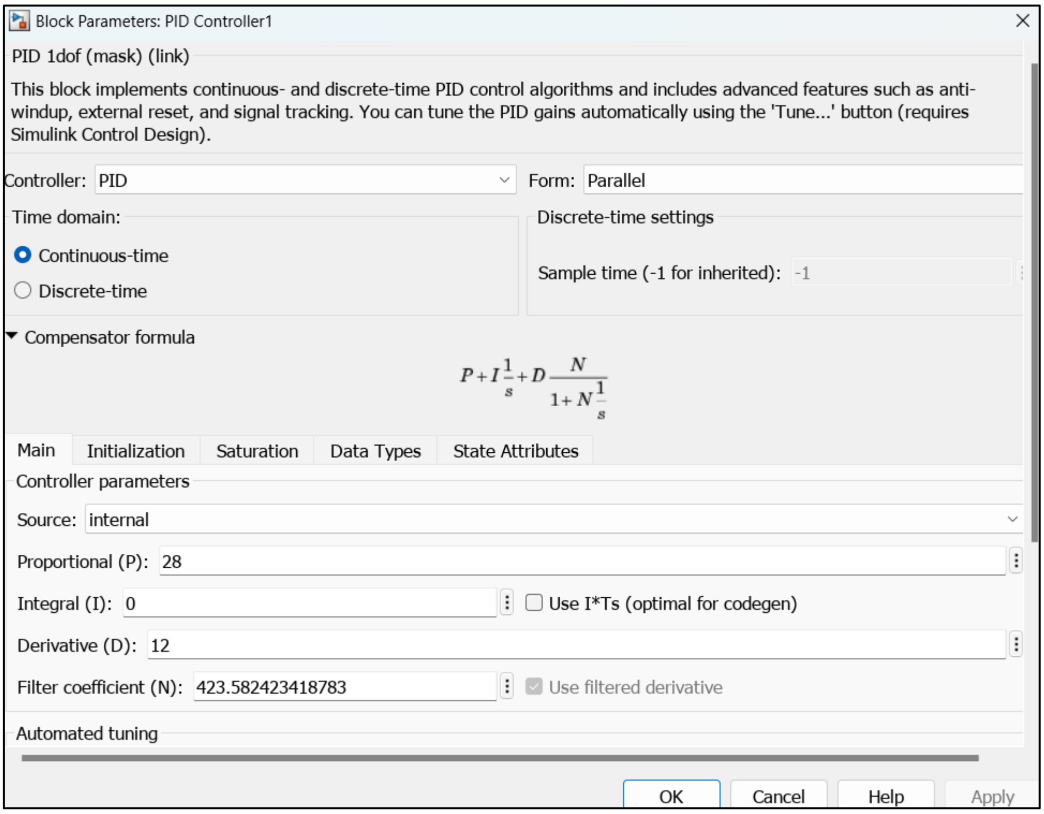
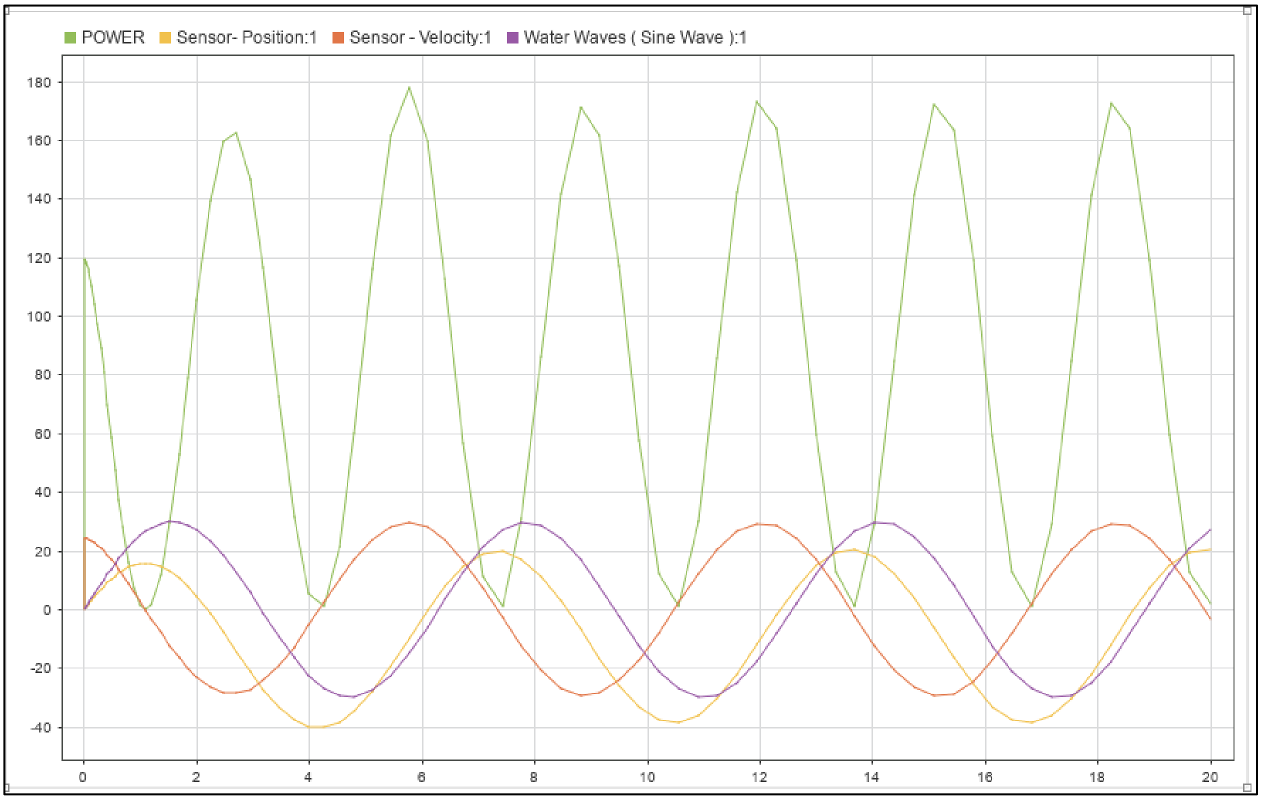


Fig - PID Parameters when auto tuned using ML modelling in simulink for first order transfer function.



Simulation results (Power, Velocity , Position) in ML tuned PID conditions ,

Kp =28 , Ki = 0 , Kd = 12 .

```
% Define PID gain ranges
kp_range = linspace(1, 50, 10);
ki_range = linspace(0, 10, 5);
kd_range = linspace(0, 10, 5);

% Initialize dataset
dataset = [];

% Loop over all combinations
for kp = kp_range
    for ki = ki_range
        for kd = kd_range
            % Assign PID values to base workspace
            assignin('base', 'Kp', kp);
            assignin('base', 'Ki', ki);
            assignin('base', 'Kd', kd);

            % Simulate the model
            simOut = sim('DL_Project', ...  % <-- change if model name is different
                'StopTime', '20', ...
                'SaveOutput', 'on', ...
                'SaveFormat', 'Dataset', ...
                'ReturnWorkspaceOutputs', 'on');

            % Extract 'logsout' data
            logsout = simOut.get('logsout');
            power_data = [];

            % Search for 'POWER' signal
            for i = 1 : logsout.numElements
                signal = logsout.get(i);
                if strcmp(signal.Name, 'POWER')
                    power_data = signal.Values.Data;
                    break;
                end
            end

            % Compute average power
            if ~isempty(power_data)
                avg_power = mean(power_data);
            else
                warning('POWER signal not found for Kp=%.2f, Ki=%.2f, Kd=%.2f. Using NaN.', kp, ki, kc
                avg_power = NaN;
            end

            % Append to dataset
            dataset = [dataset; kp, ki, kd, avg_power];
        end
    end
end

% Remove NaN rows
dataset_clean = dataset(~isnan(dataset(:, 4)), :);

% Write to CSV
csvwrite('pid_power_dataset.csv', dataset_clean);

disp('✅ CSV file "pid_power_dataset.csv" generated successfully.');
```

Fig -MATLAB Code for pid_power_dataset creation

10. Conclusion

This project successfully demonstrated the design and optimization of a Wave Energy Converter (WEC) system modeled as a mass-spring-damper system controlled by a PID controller. Through Simulink and Simscape simulations, the system was analyzed under varying PID parameters to maximize power output from ocean wave energy. Traditional PID tuning methods were compared with an advanced machine learning-based approach, which leveraged a neural network trained on simulation data to efficiently predict optimal PID gains.

The machine learning model enabled rapid exploration of a large parameter space, resulting in an optimized PID controller with gains Kp = 28 , Ki = 0 , and Kd = 12, which achieved a maximum power output of 180 W. This significantly outperformed initial tuning and Simulink's built-in PID tuner, which yielded 30 W and 160 W respectively.

Overall, this integration of vibration control, system modelling, and machine learning techniques illustrates a promising approach to maximizing energy harvesting efficiency in wave energy converters. The methodology reduces the time and computational resources required for controller tuning, enabling adaptive and efficient renewable energy solutions.

Future work may extend this framework by incorporating reinforcement learning for adaptive control, real-time implementation, and experimental validation to further enhance performance in dynamic ocean environments.

- **Find the Files attached related to the Report :**
  Project Link - DL\DL_Project.slx
  Excel sheet link - pid_power_dataset.csv.xlsx
  Code link -data_collector.m

11. References

1. MathWorks Documentation: Simulink PID Controller
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
3. A. L. Dimeas et al., "Machine Learning Based Tuning of PID Controllers," IEEE Transactions on Industrial Informatics, 2019.